EASY · ADVANCED MATH

Advanced Math

10

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ADVANCED MATH

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

A. $4ts16s^2 - 14ts$

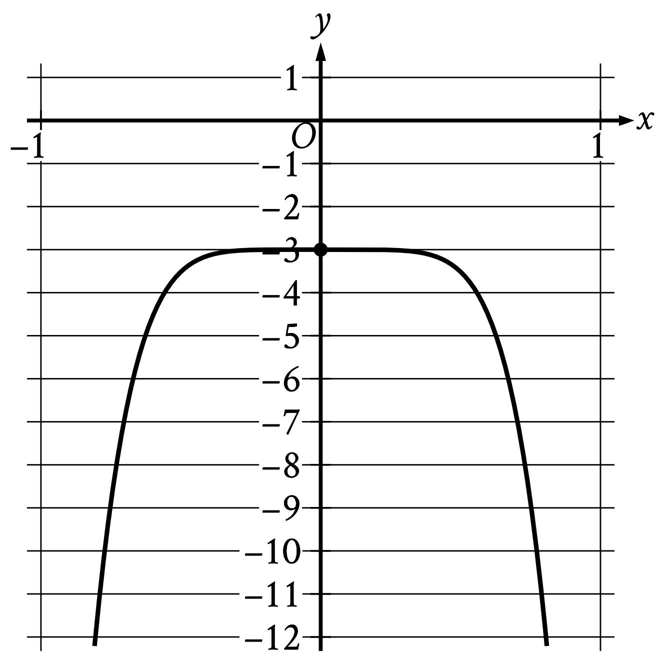
B. $4ts16t^2s^2 - 14t$

C. $4t^2s16ts - 14s$

D. $4t^2s16s^2 - 14t$

Q2

GRID-IN ADVANCED MATH



- In quadrant 3, the curve rises to its maximum at point (0 comma negative 3).
- In quadrant 4, the curve then falls.

The graph of the polynomial function f, where $y = fx$, is shown. The y-intercept of the graph is $0, y$. What is the value of y?

STUDENT-PRODUCED RESPONSE

. / . / . .

0 0 0 0
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
5 5 5 5
6 6 6 6

7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ADVANCED MATH

$$x = 3$$

$$y = 15 - x^2$$

A solution to the given system of equations is x, y. What is the value of xy?

A. 432

B. 54

C. 45

D. 18

Q4

ADVANCED MATH

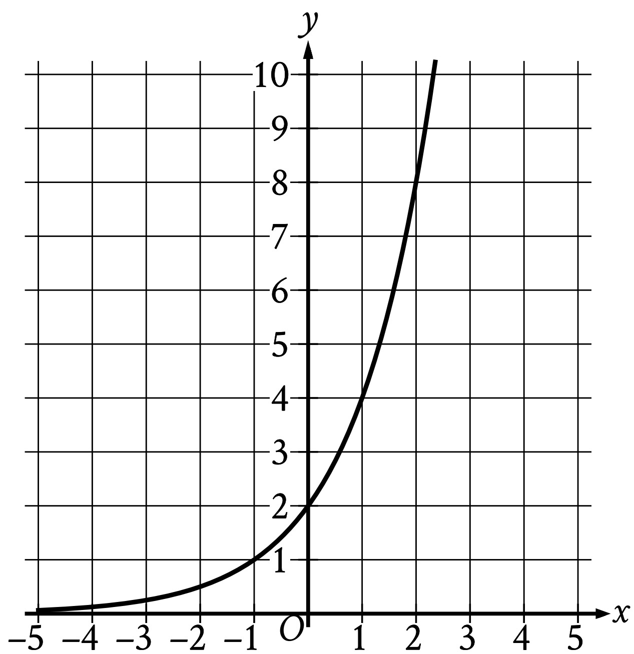
Which expression is equivalent to $12x + 27$?

A. $129x + 1$

B. $2712x + 1$

C. $34x + 9$

D. $39x + 24$



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up gradually to point (0 comma 2).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - (negative 1 comma 1)
  - (0 comma 2)
  - (2 comma 8)

What is the y-intercept of the graph shown?

A. $0, 0$

B. $0, 2$

C. $2, 0$

D. $2, 2$

Q6 ADVANCED MATH

$$c - 7 = 25p + k$$

The given equation relates the positive numbers c, p, and k. Which equation correctly expresses c in terms of p and k?

A. $c = 25p + k + 7$

B. $c = 25p + k - 7$

C. $c = 725p + k$

D. $c = \frac{25p + k}{7}$

Q7

ADVANCED MATH

Which expression is equivalent to $8yzy7z$?

A. $56y^2z^2$

B. $56y^2z$

C. $56yz$

D. $16yz$

Q8

ADVANCED MATH

The function g is defined by $gx = \sqrt{8x+1}$. What is the value of $g3$?

A. $\frac{5}{8}$

B. $\frac{25}{8}$

C. 5

D. 25

Q9 ADVANCED MATH

$$x = 8$$

$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point x, y in the xy-plane. What is the value of y?

A. 8

B. 24

C. 64

D. 72

Q10 ADVANCED MATH

Which expression is equivalent to $23x^3 + 2x^2 + 9x$?

A. $23xx^2 + 2x + 9$

B. $9x23x^3 + 2x^2 + 1$

C. $x23x^2 + 2x + 9$

D. $34x^3 + x^2 + x$